Baseline (late fusion)
config: joint
GM map
Voxelwise, 0–1
CSF map
Voxelwise, 0–1
fALFF map
Voxelwise
sFNC
1,378 edges
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
MLP encoder
217 → 128 → 64
MLP encoder
217 → 128 → 64
MLP encoder
217 → 128 → 64
MLP encoder
1,378 → 128 → 64
Average
Mean of four 64-d vectors
Head MLP
64 → 128 → 1
MDD logit
meanMLP (early fusion)
config: meanmlp
GM map
Voxelwise, 0–1
CSF map
Voxelwise, 0–1
fALFF map
Voxelwise
sFNC
1,378 edges
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
Concatenate
One 2,029-value vector
Trunk MLP
2,029 → 128 → 64
Linear
64 → 1
MDD logit
Panel A (transparent)
config: unified
GM map
Voxelwise, 0–1
CSF map
Voxelwise, 0–1
fALFF map
Voxelwise
sFNC
1,378 edges
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
Atlas means
200 + 17 regions
Linear map
217 → 53×8
Linear map
217 → 53×8
Linear map
217 → 53×8
Per-row linear
53×53 rows → 53×8
Node features H
53 nodes × 32 channels
Adjacency Â
Learned 53×53
Graph conv layer
ReLU(Â · H W), dropout
Flatten
53 × 64 = 3,392
Head MLP
3,392 → 128 → 1
Decoder
→ 128 → 4 inputs
MDD logit
Reconstructions
Error vs inputs
Panel A: reconstruction loss
decoder branch, training only
Graph output
3,392 values
GM head
→ 128 → 217
CSF head
→ 128 → 217
fALFF head
→ 128 → 217
sFNC head
→ 128 → 1,378
Squared error
mean over 217
Squared error
mean over 217
Squared error
mean over 217
Squared error
mean over 1,378
For each subject
GM input
same subject
CSF input
same subject
fALFF input
same subject
sFNC input
same subject
Mean over subjects
Per modality, in the batch
Sum over modalities
Then × 0.1
Total loss
BCE + recon + L1
☐ Data
☐ Learned layers
☐ Where modalities combine (Panel A: also nodes)
☐ Loss computation
☐ Precomputed once, before training